

lazy when the task gets harder. He is surprised by the ease with which intelligent people appear satisfied enough with their initial answer that they stop thinking.

Kahneman tells us activities that put demands on System 2 thinking require self-control, and continuous exertion of self-control can be unpleasant. If we are continually forced to do something over and over that is challenging, there is a tendency to exert less self-control when the next challenge arrives. Eventually we simply run out of gas. In contrast, “those who avoid the sin of intellectual sloth could be called ‘engaged.’ They are more alert, more intelligently active, less willing to be satisfied with superficially attractive answers, more skeptical about their intuitions.”⁸

Shane Frederick, associate professor of marketing at Yale University, has given us a fascinating look at how people with fairly high IQs navigate between System 1 and System 2 thinking. He gathered a group of Ivy League students from Harvard, Princeton, and MIT (presumably all highly intelligent) and asked them the following three questions:

1. A bat and ball cost \$1.10. The bat costs one dollar more than the ball. How much does the ball cost?
2. If it takes five machines five minutes to make five widgets, how long would it take 100 machines to make 100 widgets?
3. In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long will it take for the patch to cover half of the lake?⁹

To Frederick’s surprise, over half of the students got the answers wrong, leading him to delineate two significant problems. First, people are not accustomed to thinking hard about problems and often rush to the first plausible answer that comes to mind so they don’t have to engage the heavy burden of System 2 thinking. The second problem, which is disturbing in itself, was the realization that the System 2 process does a very poor job of monitoring System 1 thinking for errors. It seemed clear to Frederick that the students were stuck in System 1 thinking and could not, or would not, convert to System 2.

How do System 1 and System 2 thinking work in investing? Say an investor is considering making a common stock purchase. Using System 1 thinking, the investor would tabulate a company’s price-to-earnings ratio,